Programming Assignment 4: BST

DISCUSSION

Comments:

My program is functioning as it is designed to do in accordance with the problem description. The driver program performs all the operations: Insert node(s), Traverse Preorder, Search BST, Delete node, Leaf Count, Sibling of a node, and Quit.

I implemented the program to use a while loop to let the user repeatedly run the program in the form of a menu until the user wants to quit out of the program.

Corner Cases:

I have tested my driver program with the examples provided by the problem description, but I have also encountered an issue in that when the expected user input is an integer (i.e. profit, tip) or has a space (i.e. name), the program infinitely loops and prints to the terminal each loop.

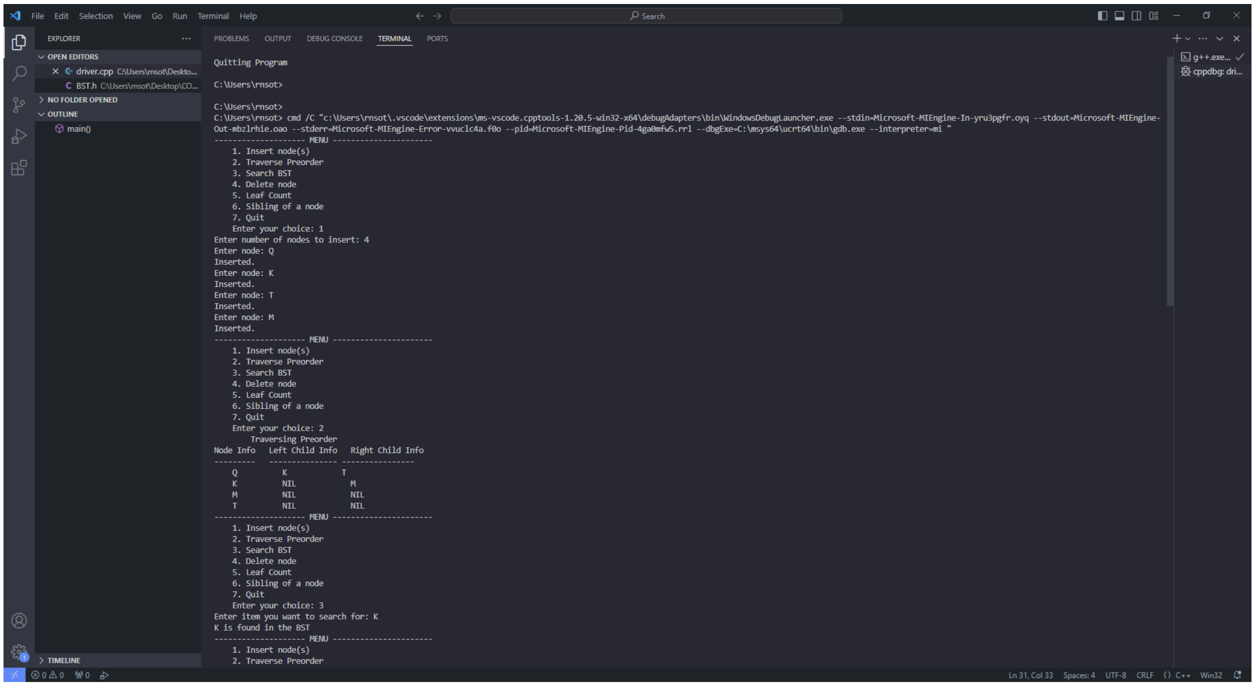
This isn't too much of a problem, but it is an error with specific inputs where the program will cease to work.

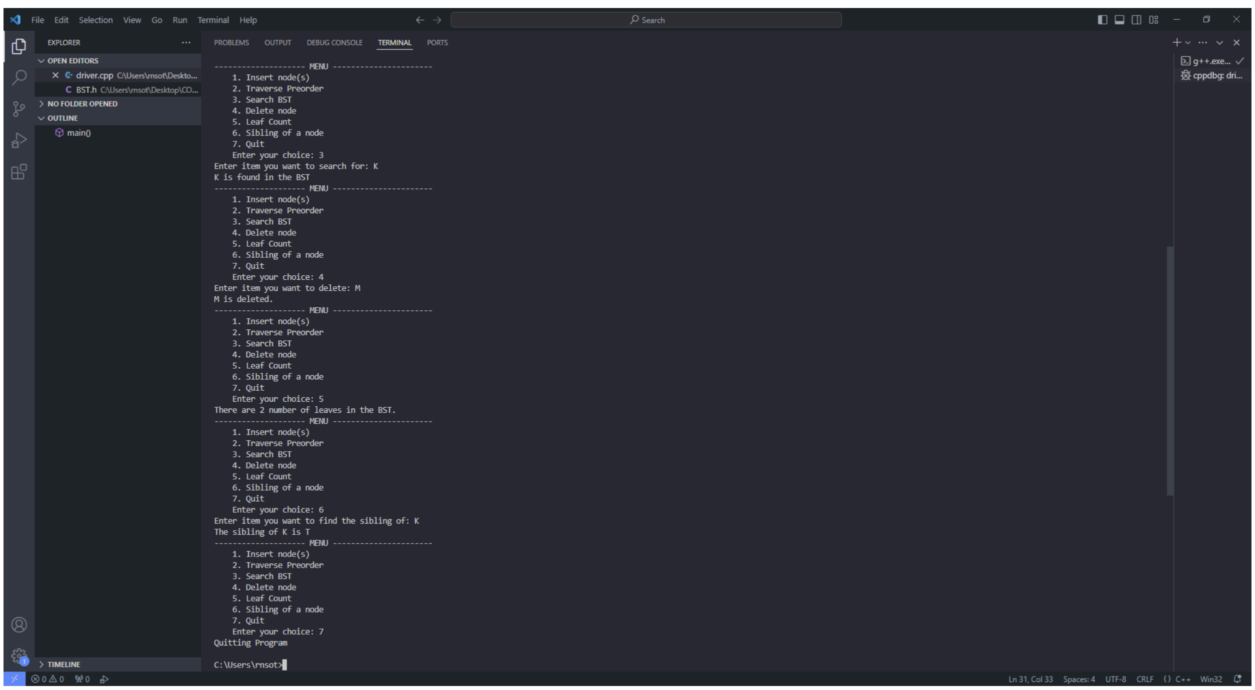
I have not tested:

- Extremely large positive values for inputting nodes
- Inputting invalid data types aside from strings/chars for ints

I have included the screenshots as well as the transcript of the runtime output of the program.

[Screenshots of Runtime Output of Code]





[Transcript of Runtime Output of Code]

```
C:\Users\rnsot>
C:\Users\rnsot> cmd /C
"c:\Users\rnsot\.vscode\extensions\ms-vscode.cpptools-1.20.5-win32-x64\debugAdapters\bin\Wi
ndowsDebugLauncher.exe --stdin=Microsoft-MIEngine-In-yru3pgfr.oyq
--stdout=Microsoft-MIEngine-Out-mbzlrhie.oao --stderr=Microsoft-MIEngine-Error-vvuclc4a.f0o
--pid=Microsoft-MIEngine-Pid-4ga0mfw5.rrl --dbgExe=C:\msys64\ucrt64\bin\gdb.exe
--interpreter=mi "
-------------------- MENU ----------------------
   1. Insert node(s)
   2. Traverse Preorder
   3. Search BST
   4. Delete node
   5. Leaf Count
   6. Sibling of a node
   7. Quit
   Enter your choice: 1
Enter number of nodes to insert: 4
Enter node: Q
Inserted.
Enter node: K
Inserted.
Enter node: T
Inserted.
Enter node: M
Inserted.
-------------------- MENU ----------------------
   1. Insert node(s)
   2. Traverse Preorder
   3. Search BST
   4. Delete node
   5. Leaf Count
   6. Sibling of a node
   7. Quit
   Enter your choice: 2
      Traversing Preorder
Node Info   Left Child Info   Right Child Info
---------   --------------- ----------------
   Q         K           T
   K         NIL           M
   M         NIL           NIL
   T         NIL           NIL
-------------------- MENU ----------------------
```

1. Insert node(s)
2. Traverse Preorder
3. Search BST
4. Delete node
5. Leaf Count
6. Sibling of a node
7. Quit

Enter your choice: 3

Enter item you want to search for: K
K is found in the BST
-------------------- MENU ----------------------

1. Insert node(s)
2. Traverse Preorder
3. Search BST
4. Delete node
5. Leaf Count
6. Sibling of a node
7. Quit

Enter your choice: 4

Enter item you want to delete: M
M is deleted.
-------------------- MENU ----------------------

1. Insert node(s)
2. Traverse Preorder
3. Search BST
4. Delete node
5. Leaf Count
6. Sibling of a node
7. Quit

Enter your choice: 5

There are 2 number of leaves in the BST.
-------------------- MENU ----------------------

1. Insert node(s)
2. Traverse Preorder
3. Search BST
4. Delete node
5. Leaf Count
6. Sibling of a node
7. Quit

Enter your choice: 6

Enter item you want to find the sibling of: K
The sibling of K is T
-------------------- MENU ----------------------

1. Insert node(s)

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   2. Traverse Preorder
   3. Search BST
   4. Delete node
   5. Leaf Count
   6. Sibling of a node
   7. Quit
   Enter your choice: 7
Quitting Program

C:\Users\rnsot>
```
